3 Convolutional neural networks

- for image data, each example consists of a two-dimensional *grid of pixels*
- depending on whether we are handling black-and-white or color images, each pixel location might be associated with either one or multiple numerical values, respectively
- one way of dealing with this rich structure is to simply discard each image's spatial structure by flattening them into one-dimensional vectors, and feeding them through a fully-connected MLP, which is deeply unsatisfying
- because these networks are invariant to the order of the features, we could get similar results regardless of whether we preserve an order corresponding to the spatial structure of the pixels, or if we permute the columns of our design matrix before fitting the MLP's parameters
- preferably, we would leverage our prior knowledge that nearby pixels are typically related to each other, to build efficient models for learning from image data

3 Convolutional neural networks

- in this chapter, we introduce *convolutional neural networks* (*CNNs*), a powerful family of neural networks that are designed for precisely this purpose
- CNN-based architectures are now ubiquitous in the field of computer vision, and have become so dominant that hardly anyone today would develop a commercial application or enter a competition related to image recognition, object detection, or semantic segmentation, without building off of this approach

3 Convolutional neural networks

- modern CNNs owe their design to inspirations from biology, group theory, and much experimentation
- in addition to their sample efficiency in achieving accurate models, CNNs tend to be computationally efficient, both because they require fewer parameters than fully-connected architectures, and because convolutions are easy to parallelize across GPU cores
- consequently, CNNs are often applied whenever possible, and increasingly they have emerged as good competitors even on tasks with a one-dimensional sequence structure, such as audio, text, and time series analysis, where recurrent neural networks are conventionally used
- some clever adaptations of CNNs have also brought applications on graph-structured data and in recommender systems

3 Convolutional neural networks

- first, we will walk through the basic operations that comprise the backbone of all convolutional networks
- these include the convolutional layers themselves, details including padding and stride, the pooling layers used to aggregate information across adjacent spatial regions, the use of multiple channels at each layer, and a careful discussion of the structure of modern architectures
- we will discuss the example of LeNet, the first convolutional network successfully deployed, long before the rise of modern deep learning
- then, we will dive into full details of some popular and comparatively recent CNN architectures, whose designs represent most of the techniques commonly used today